

SANTIAGO DEL ESTERO, Argentina

WMO: 871290

Lat: 27.7547S

Lon: 64.2997W

Elev: 199

StdP: 98.96

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-0.6	1.6	-7.6	2.0	7.3	-5.6	2.4	8.7	9.9	18.1	8.9	20.9	0.8	320	0.363

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.1	39.2	23.6	37.6	23.5	36.1	23.1	26.4	33.7	25.7	33.1	25.0	32.3	4.7	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.6	20.0	30.2	23.8	19.1	29.1	23.1	18.3	28.3	84.1	33.9	80.5	33.1	77.6	32.4	30.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.2	8.0	6.5	-4.0	42.0	1.7	1.3	-5.2	43.0	-6.2	43.7	-7.2	44.4	-8.4	45.4		
(5)			-5.1	27.7	1.7	0.9	-6.3	28.4	-7.3	28.9	-8.2	29.4	-9.5	30.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	20.9	27.3	26.2	24.1	20.6	16.7	13.6	13.2	15.9	19.1	23.0	25.0	26.8	(6)
(7)		DBStd	6.27	3.21	3.50	3.31	3.85	3.84	3.48	4.05	4.60	4.62	4.14	3.78	3.45	(7)
(8)		HDD10.0	27	0	0	0	0	1	8	14	4	1	0	0	0	(8)
(9)		HDD18.3	571	0	1	2	20	78	145	167	105	44	7	2	0	(9)
(10)		CDD10.0	4017	537	453	436	317	210	117	112	186	275	402	451	521	(10)
(11)		CDD18.3	1520	279	220	179	88	29	4	7	29	69	152	202	262	(11)
(12)		CDH23.3	17151	3343	2397	1678	712	204	33	111	449	910	1756	2394	3162	(12)
(13)	CDH26.7	8508	1816	1204	733	249	46	3	27	185	398	874	1254	1718	(13)	
(14)	Wind	WSAvg	2.6	2.6	2.5	2.3	2.3	2.1	2.0	2.4	2.7	3.1	3.2	3.1	2.9	(14)
(15)	Precipitation	PrecAvg	593	123	104	87	36	14	7	3	11	43	68	97		(15)
(16)		PrecMax	1163	323	523	202	188	61	47	22	10	72	212	256	298	(16)
(17)		PrecMin	282	18	7	12	0	0	0	0	0	0	1	1	5	(17)
(18)		PrecStd	160	68	78	49	34	15	9	5	2	14	44	46	57	(18)
(19)	Monthly Design Dry Bulb and	DB	40.7	39.7	37.4	34.1	30.4	27.0	30.1	35.5	37.9	39.9	40.4	41.4	(19)	
(20)		MCWB	24.2	24.1	24.2	23.1	21.5	19.7	18.4	20.0	20.6	22.7	23.7	23.5	(20)	
(21)		DB	38.3	37.1	34.7	31.7	27.8	23.9	26.1	31.3	33.9	36.8	37.5	38.8	(21)	
(22)		MCWB	24.2	24.5	24.0	22.3	20.3	17.2	17.3	18.8	19.6	22.0	22.5	23.8	(22)	
(23)	Mean Coincident Wet Bulb Temperatures	DB	36.6	35.2	32.8	29.5	25.6	22.0	23.8	28.1	31.2	34.1	35.5	36.6	(23)	
(24)		MCWB	23.8	24.3	23.5	22.0	19.1	15.7	15.6	17.0	18.3	20.8	21.9	23.5	(24)	
(25)		DB	34.7	33.2	30.8	27.3	23.5	20.2	21.3	25.2	28.6	31.6	33.3	34.5	(25)	
(26)		MCWB	23.6	23.7	22.7	20.8	18.0	14.8	13.7	15.4	17.2	20.0	21.4	23.1	(26)	
(27)	Monthly Design Wet Bulb and	WB	27.2	27.4	26.5	25.2	22.7	20.4	19.7	21.1	22.6	24.9	26.7	27.1	(27)	
(28)		MCDB	34.5	34.7	32.9	30.9	28.3	24.9	28.0	32.0	34.0	35.8	33.2	35.2	(28)	
(29)		WB	26.2	26.1	25.3	24.0	21.5	18.9	18.1	19.4	21.0	23.5	24.7	25.7	(29)	
(30)		MCDB	33.9	33.0	31.7	29.1	25.6	22.4	24.8	30.0	30.6	33.3	33.2	34.4	(30)	
(31)	Mean Coincident Dry Bulb Temperatures	WB	25.3	25.4	24.6	22.9	20.4	17.5	16.8	18.0	19.7	22.3	23.6	24.9	(31)	
(32)		MCDB	33.0	32.1	30.5	27.3	23.9	20.2	22.2	26.8	28.9	30.9	32.3	33.1	(32)	
(33)		WB	24.6	24.6	23.7	21.9	19.3	16.2	15.2	16.2	18.3	21.1	22.5	24.1	(33)	
(34)		MCDB	31.9	31.0	28.8	25.8	22.7	18.8	19.5	23.8	26.6	29.1	30.5	31.8	(34)	
(35)	Mean Daily Temperature Range	MDBR	12.1	11.2	10.6	10.6	10.8	11.7	14.1	15.5	15.1	13.1	13.0	12.4	(35)	
(36)		MCDBR	14.9	14.1	13.9	13.7	13.7	14.8	17.6	19.6	18.7	17.2	16.2	15.9	(36)	
(37)		MCWBR	4.8	4.5	5.4	6.3	7.1	8.8	10.0	9.6	8.4	6.8	5.9	5.3	(37)	
(38)		MCDBR	12.4	11.7	11.0	10.4	10.5	10.2	14.9	17.9	15.8	14.4	13.1	12.9	(38)	
(39)		MCWBR	4.9	4.7	4.9	5.5	5.7	6.7	8.8	8.9	7.6	6.2	5.6	5.3	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.410	0.407	0.393	0.380	0.354	0.339	0.341	0.396	0.462	0.442	0.411	0.412	(40)	
(41)		taud	2.396	2.405	2.443	2.436	2.461	2.481	2.441	2.267	2.077	2.219	2.339	2.375	(41)	
(42)		Ebn at Noon	934	919	901	862	840	833	845	831	820	876	926	934	(42)	
(43)		Edh at Noon	128	123	113	103	91	85	91	120	160	147	134	131	(43)	
(44)	All-Sky Solar Radiation	RadAvg	6.70	6.06	4.85	3.82	3.08	2.79	3.37	4.34	5.25	5.86	6.59	6.65	(44)	
(45)		RadStd	0.50	0.43	0.49	0.43	0.39	0.35	0.28	0.31	0.49	0.57	0.59	0.41	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	+0.50	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	N/A	N/A	N/A	+0.82	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page